AUDIT REPORT

order-outcome-margin: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/order-outcome-margin
Numbers taken from commit	`e78d53a391889af780be9f07f440398e04edc426` (outputs last changed in `e78d53a (2026-10-13 22:05:19 +0400)`)
Video	`docs/demo/demo.mp4` · 88.4 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 19:03
Automated checks	60 pass · 0 fail · 1 manual (of 61 figures)
Assumptions	12 (A-01 to A-12)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # data/, outputs/*.csv, results.json, report.md, *.png (about 2 min)
python3 docs/demo/source/tools/export_run.py outputs docs/demo/source/assets/run.json
python3 docs/demo/source/tools/inject_block.py docs/demo/source/assets/run.json docs/demo/source/compositions/s04-routes.html --block RUN --keys daily,frontier,points
python3 docs/demo/source/tools/make_end_plot.py outputs docs/demo/source/assets/end-frontier.png
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_cod_sheet.py docs/demo/source/assets/cod_sheet.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/cod_sheet.json docs/demo/source/compositions/value.html --rows 315400,316055,316096,316229,315744,316309 --animate 315400,316096,316229,315744
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/order-outcome-margin-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/order-outcome-margin-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 chips	01	author's framing; README 'The problem' (finance and operations teams)	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem'	README lines 12-25	-	general knowledge	✓ verified
F-03	341,880 orders	02, 05	outputs/results.json	orders_total	-	direct	✓ verified
F-04	2 years / 2024–2025	02, 05	src/order_margin/config.py:70-71	DATA_START, DATA_END	-	direct	✓ verified
F-05	48%	02	data/orders.csv.gz	payment == 'cod'	share of COD orders = 48.3%	derived	✓ verified
F-06	Five ways each one can end / 5 ways it could end	02, 03	src/order_margin/config.py:11	CLASSES	len = 5	direct	✓ verified
F-07	~470 orders a day	02	data/orders.csv.gz	orders / distinct order days	341,880 / 731 = 467.7, rounded to tens (README: 'around 470 orders a day')	derived	✓ verified
F-08	6 delivery zones, 2 remote	02	src/order_margin/config.py:17-18	ZONES, REMOTE_ZONES	-	direct	✓ verified
F-09	1 in 5 orders is not delivered	02	data/orders.csv.gz	outcome != delivered	1 - 0.798 = 20.2% = 1 in 5 (test window: 20.9%)	derived	✓ verified
F-10	scored as cash and as card	03	src/order_margin/pipeline.py:193-201	with_payment(Xt_cod, 'card')	-	direct	✓ verified
F-11	daily margin / cash rules / beats risk rule	03	outputs/daily_margin.csv, outputs/results.json	122 daily rows; margin_minus_risk_rule > 0	-	derived	✓ verified
F-12	CALIBRATE · make the odds honest	03	src/order_margin/calibration.py:26-39	IsotonicCalibrator	-	direct	✓ verified
F-13	+22,258	04, 06, 08, 09	outputs/policy.csv	policy='margin rule, gbm + isotonic', margin_change	-	direct	✓ verified
F-14	−2,113	04, 06, 08	outputs/policy.csv	policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	direct	✓ verified
F-15	config.py, generate.py, features.py, models.py, calibration.py, economics.py, policy.py, pipeline.py, data/truth	05	src/order_margin/	classes and functions	-	direct	✓ verified
F-16	23 columns	05	src/order_margin/features.py:21-45	NUMERIC (20) + CATEGORICAL (3)	20 + 3 = 23	direct	✓ verified
F-17	history ≥14 d old	05	src/order_margin/features.py:15	RESOLVE_DAYS = 14	-	direct	✓ verified
F-18	186 boosting rounds	05	outputs/results.json	gbm_rounds	-	direct	✓ verified
F-19	5 outcomes at once	05	src/order_margin/models.py:74-75	objective multiclass, num_class = N_CLASSES = 5	-	direct	✓ verified
F-20	fit May–Aug 25	05	src/order_margin/config.py:68-69	EARLY_STOP_END 2025-04-30, CALIB_END 2025-08-31	-	direct	✓ verified
F-21	40% / 60% · assumed input	05	src/order_margin/config.py:75	COD_TO_CARD_CONVERSION	-	direct	✓ verified
F-22	72.8 s	05	outputs/results.json	runtime_seconds	-	direct	✓ verified
F-23	73,024 orders	05, 08, 09	outputs/results.json	orders_test	-	direct	✓ verified
F-24	how well the odds fit · daily margin error · COD rules scored on the true odds · 95% CI	05	src/order_margin/pipeline.py:139-160, 274-275; calibration.py:64-83	-	-	direct	✓ verified
F-25	4 date windows · train · tune · calibrate · test	05	src/order_margin/pipeline.py:64-71	split_masks keys	-	direct	✓ verified
F-26	SVG line chart	06	outputs/daily_margin.csv via tools/export_run.py → assets/run.json → RUN block in s04-routes.html	realized, class priors, gbm + isotonic, gbm class-weighted, raw, peak	values rounded to whole units; drawn, not retyped	direct	✓ verified
F-27	122 test days	06	outputs/results.json	test_days	-	direct	✓ verified
F-28	22.5%	06, 08	outputs/metrics.csv	model='class priors', daily_mape_peak (0.2254)	x 100	direct	✓ verified
F-29	4.3%	06, 08	outputs/metrics.csv	model='gbm + isotonic', daily_mape_peak (0.0431)	x 100	direct	✓ verified
F-30	Class weighting forecasts a loss every	06	outputs/daily_margin.csv	column 'gbm class-weighted, raw' max =	max < 0	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
	day.			-10,259			
F-31	SVG line chart, y clipped to -60k..+40k	06	outputs/policy_frontier.csv, outputs/policy.csv via tools/export_run.py	orders_lost, margin_change per ranking; policy rows	same symlog x-axis as plots.py (linear below 100)	direct	✓ verified
F-32	−509,868	06	outputs/policy.csv	policy='block every COD order', margin_change	-	direct	✓ verified
F-33	14,469 orders lost	06	outputs/policy.csv	orders_lost = 14,468.6	rounded	direct	✓ verified
F-34	hidden on 15.5% of COD orders	06	outputs/policy.csv	policy='margin rule, gbm + isotonic', blocked_share_of_cod (0.1554)	x 100	direct	✓ verified
F-35	35,389 COD orders	06, 08, 09	outputs/results.json	cod_orders_test	-	direct	✓ verified
F-36	Risk score · same block rate	06	src/order_margin/pipeline.py:206-218	threshold_for_share(risk_score(pc_c), share)	-	direct	✓ verified
F-37	SALE WEEK · 2025-11-24	07	demo assets/cod_sheet.json; src/order_margin/generate.py:30	day, is_peak	-	direct	✓ verified
F-38	609 COD orders (6 rows + 603 more)	07	demo assets/cod_sheet.json; data/orders.csv.gz	cod_orders; COD orders placed on 2025-11-24	-	direct	✓ verified
F-39	102 of 609 COD orders hide COD · 507 keep it	07	demo assets/cod_sheet.json	blocked, kept	102 + 507 = 609	direct	✓ verified
F-40	+602 margin that day	07	demo assets/cod_sheet.json	day_true_gain	sum over hidden orders of conv x E_true[card] - E_true[cod]	direct	✓ verified
F-41	315400, 316055, 316096, 316229, 315744, 316309	07	demo assets/cod_sheet.json, injected into compositions/value.html	rows	gain = conv x IF CARD - IF COD; action = HIDE COD if gain > 0	direct	✓ verified
F-42	(consistency check, not on screen)	07	demo assets/cod_sheet.json; outputs/policy.csv	test_window_blocked vs policy 'margin rule, gbm + isotonic' cod_orders_blocked	5,498 = 5,498; 186 rounds = 186	derived	✓ verified
F-43	+ 603 more COD orders	07	F-38	609 - 6 rows	609 - 6 = 603	derived	✓ verified
F-44	63%	08	outputs/policy.csv	margin rule 22,258 / true-probability ceiling 35,479	22,258 / 35,479 = 62.7%	derived	✓ verified
F-45	perfect model: +35,479	08	outputs/policy.csv	policy='margin rule, true probabilities (ceiling)', margin_change	-	direct	✓ verified
F-46	+24,371	08	outputs/results.json	margin_minus_risk_rule	-	direct	✓ verified
F-47	95% CI: 21,580–27,556	08	outputs/results.json	margin_minus_risk_rule_ci95	-	direct	✓ verified
F-48	16.47×	09	outputs/metrics.csv	model='gbm class-weighted, raw', count_ratio_fraud	-	direct	✓ verified
F-49	3.80×	09	outputs/metrics.csv	model='gbm class-weighted, raw', count_ratio_failed	-	direct	✓ verified
F-50	2.49×	09	outputs/metrics.csv	model='gbm class-weighted, raw', count_ratio_returned	-	direct	✓ verified
F-51	1.12×	09	outputs/metrics.csv	model='gbm + isotonic', count_ratio_fraud	-	direct	✓ verified
F-52	1.02×	09	outputs/metrics.csv	model='gbm + isotonic', count_ratio_failed	-	direct	✓ verified
F-53	0.98×	09	outputs/metrics.csv	model='gbm + isotonic', count_ratio_returned	-	direct	✓ verified
F-54	64% of cash orders	09	outputs/policy.csv	policy='margin rule, class-weighted gbm', blocked_share_of_cod (0.6393)	x 100	direct	✓ verified
F-55	−168,782	09	outputs/policy.csv	policy='margin rule, class-weighted gbm', margin_change	-	direct	✓ verified
F-56	bars 260 / 60 / 39 px (lesson) and 18 / 16 / 15 px (fix)	09	outputs/metrics.csv	count_ratio_* / max x 260	round(ratio / 16.4749 x 260)	derived	✓ verified
F-61	Ship the plain model, calibrated · PICKED · the weighted one is only a comparison	09	src/order_margin/models.py:71; pipeline.py:122, 180, 203, 229, 252; tests/test_pipeline.py:24	-	-	direct	✓ verified
F-57	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-58	about 2 min	10	README.md line 137	'about 2 minutes on a laptop'	-	direct	✓ verified
F-59	LightGBM · scikit-learn · pandas · Python	10	pyproject.toml dependencies	lightgbm, scikit-learn, pandas	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-60	$ make run	10	Makefile	run target	-	direct	✓ verified

2. Automated check run (2026-10-14 19:03)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 chips	-	manual
F-02	true	true	✓ pass
F-03	341,880	341,880	✓ pass
F-04	true	true	✓ pass
F-05	48%	48%	✓ pass
F-06	5	5	✓ pass
F-07	~470	~470	✓ pass
F-08	true	true	✓ pass
F-09	1 in 5	1 in 5	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	+22,258	+22,258	✓ pass
F-14	-2,113	-2,113	✓ pass
F-15	true	true	✓ pass
F-16	23	23	✓ pass
F-17	true	true	✓ pass
F-18	186	186	✓ pass
F-19	true	true	✓ pass
F-20	true	true	✓ pass
F-21	true	true	✓ pass
F-22	72.8 s	72.8 s	✓ pass
F-23	73,024	73,024	✓ pass
F-24	true	true	✓ pass
F-25	true	true	✓ pass
F-26	true	true	✓ pass
F-27	122	122	✓ pass
F-28	22.5%	22.5%	✓ pass
F-29	4.3%	4.3%	✓ pass
F-30	true	true	✓ pass
F-31	true	true	✓ pass
F-32	-509,868	-509,868	✓ pass
F-33	14,469	14,469	✓ pass
F-34	15.5%	15.5%	✓ pass
F-35	35,389	35,389	✓ pass
F-36	true	true	✓ pass
F-37	true	true	✓ pass
F-38	true	true	✓ pass
F-39	true	true	✓ pass
F-40	+602	+602	✓ pass
F-41	true	true	✓ pass
F-42	true	true	✓ pass
F-43	603	603	✓ pass
F-44	63%	63%	✓ pass
F-45	+35,479	+35,479	✓ pass
F-46	+24,371	+24,371	✓ pass
F-47	21,580–27,556	21,580–27,556	✓ pass
F-48	16.47×	16.47×	✓ pass
F-49	3.80×	3.80×	✓ pass
F-50	2.49×	2.49×	✓ pass
F-51	1.12×	1.12×	✓ pass
F-52	1.02×	1.02×	✓ pass
F-53	0.98×	0.98×	✓ pass
F-54	64%	64%	✓ pass
F-55	-168,782	-168,782	✓ pass
F-56	true	true	✓ pass
F-61	true	true	✓ pass
F-57	true	true	✓ pass
F-58	true	true	✓ pass
F-59	true	true	✓ pass
F-60	true	true	✓ pass

3. Assumptions register

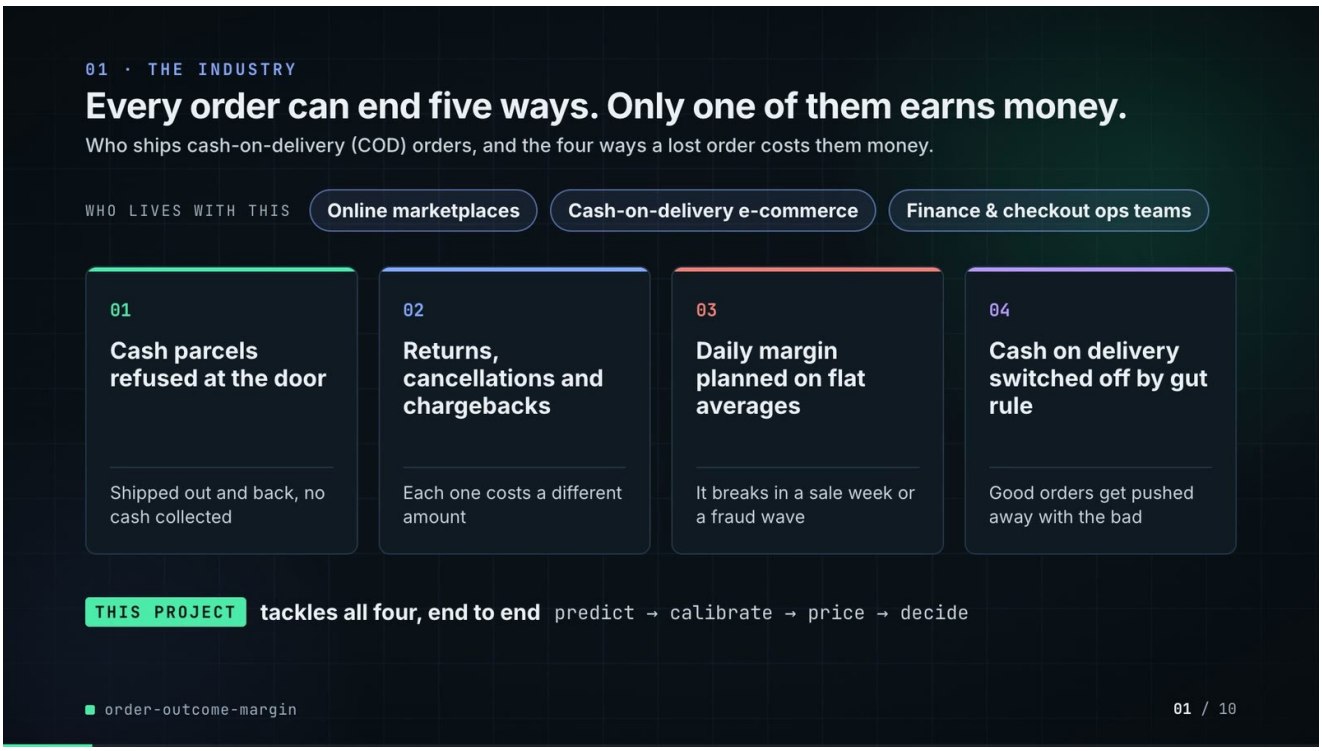
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator	src/order_margin/config.py:78 (SEED = 20260913);	all figures	slides 01, 02,

ID	Assumption	Source	Figures affected	Slides affected
	sized to the production problem (341,880 orders over 2024-2025, six zones, five categories, COD about 48%). The production data is proprietary.	src/order_margin/generate.py; README 'The data'		03, 04, 05, 06, 07, 08, 09, 10
A-02	Card conversion: 40% of blocked customers younger than 30 days and 60% of the rest switch to a card. It is an input, not measured; with decisions fixed, 20 points lower conversion breaks even (+944) and 30 points lower loses 9,714.	src/order_margin/config.py:75-76 (COD_TO_CARD_CONVERSION, NEW_CUSTOMER_DAYS); outputs/conversion_sensitivity.csv; README 'Known limitations' (biggest gap)	F-13, F-14, F-21, F-32, F-33, F-34, F-39, F-40, F-41, F-44, F-45, F-46, F-47, F-54, F-55	slides 04, 05, 06, 07, 08, 09
A-03	Unit economics are one table per category (gross margin, resale loss, shipping legs, COD/card/wallet fees, a 45 chargeback fee). Real courier rates vary by weight and contract.	src/order_margin/config.py:22-60 (Economics); src/order_margin/economics.py:36-71	F-13, F-14, F-28, F-29, F-30, F-32, F-40, F-41, F-44, F-45, F-46, F-47, F-55	slides 04, 06, 07, 08, 09
A-04	Time split: train 2024-01-01..2025-02-28, early stopping 2025-03..04, calibration 2025-05..08, test 2025-09-01..12-31 (73,024 orders, 122 days).	src/order_margin/config.py:67-71; pipeline.py:64-71 (split_masks)	F-20, F-23, F-25, F-27, F-28, F-29, F-35, F-48, F-49, F-50, F-51, F-52, F-53	slides 05, 06, 08, 09
A-05	Outcomes are known 14 days after the order; customer history features only use orders placed at least 14 days earlier, shrunk towards fixed long-run rates.	src/order_margin/features.py:15-19 (RESOLVE_DAYS, PRIOR_STRENGTH, POPULATION_RATE)	F-17	slides 05
A-06	COD policies are scored on the generator's true outcome probabilities under COD and under card (data/truth/order_truth.csv.gz), never on the model that made the decision. The card version of an order is the same order with the payment feature switched.	src/order_margin/pipeline.py:186-201 (counterfactual), :236-245 (score_policy); features.py:with_payment; README 'How it works'	F-13, F-14, F-32, F-40, F-44, F-45, F-46, F-47, F-55	slides 04, 06, 07, 08, 09
A-07	LightGBM settings: multiclass, learning rate 0.03, 15 leaves, seed 7, early stopping with 100-round patience on the March-April 2025 window (stopped at 186 rounds); chosen from four settings, never on the test window.	src/order_margin/models.py:71-108; README 'How it works'	F-18, F-19, F-48, F-49, F-50, F-51, F-52, F-53	slides 05, 09
A-08	One seed and one test window. The bootstrap interval (1,000 order resamples) covers which orders arrived, not a different year or fraud pattern.	src/order_margin/pipeline.py:268-275; README 'Known limitations'	F-46, F-47	slides 08
A-09	'Sale days' = days flagged is_peak: the November sale week (2025-11-21..29) plus 11.11. Slide 06 shades all ten of them (label 'SALE DAYS (shaded)').	src/order_margin/generate.py:30-31 (SALE_WEEKS, SINGLES_DAY); pipeline.py:74-85 (_daily_errors)	F-28, F-29, F-37, F-26	slides 06, 07, 08
A-10	Video rounding: margins to whole units with thousands separators, percentages to 1 decimal (63%, 48% and 64% to whole numbers), predicted/actual count ratios to 2 decimals; chart axes in thousands ('+20k'); '~470 a day' and 'about 2 min' are rounded labels.	this audit (display convention)	F-05, F-07, F-09, F-22, F-28, F-29, F-34, F-44, F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-58	slides 02, 05, 06, 08, 09, 10
A-11	Slide 01 (industry, everyday problems) is general e-commerce knowledge used as context, matched to the README problem statement. It is not a result of this repo and carries no numbers.	author's framing; README 'The problem'	F-01, F-02	slides 01
A-12	The shipped model is the plain (unweighted) LightGBM with an isotonic calibrator fitted on May-Aug 2025, a window the booster never trained or early-stopped on. The COD rule and the reported daily forecast use it; the class-weighted LightGBM is fitted only as a comparison, and a test requires that weighting breaks the margin forecast.	src/order_margin/models.py:71 (class_weighted: bool = False); pipeline.py:122 (isotonic fitted on the calibration window), :180 (daily forecast columns), :203 (g_iso = gain_hat(gbm, iso['gbm'])), :205 and :229 (weighted model as a scored comparison), :252 (block_iso); tests/test_pipeline.py:24	F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-55, F-61	slides 09

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every order can end five ways. Only one of them earns money.

Who ships cash-on-delivery (COD) orders, and the four ways a lost order costs them money.

WHO LIVES WITH THIS · Online marketplaces · Cash-on-delivery e-commerce · Finance & checkout ops teams

01 Cash parcels refused at the door — Shipped out and back, no cash collected

02 Returns, cancellations and chargebacks — Each one costs a different amount

03 Daily margin planned on flat averages — It breaks in a sale week or a fraud wave

04 Cash on delivery switched off by gut rule — Good orders get pushed away with the bad

THIS PROJECT tackles all four, end to end · predict → calibrate → price → decide

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 chips	`author's framing; README 'The problem' (finance and operations teams)·-`	-	manual
F-02	4 problem cards	`README 'The problem'` · README lines 12-25	-	✓

Assumptions: A-11

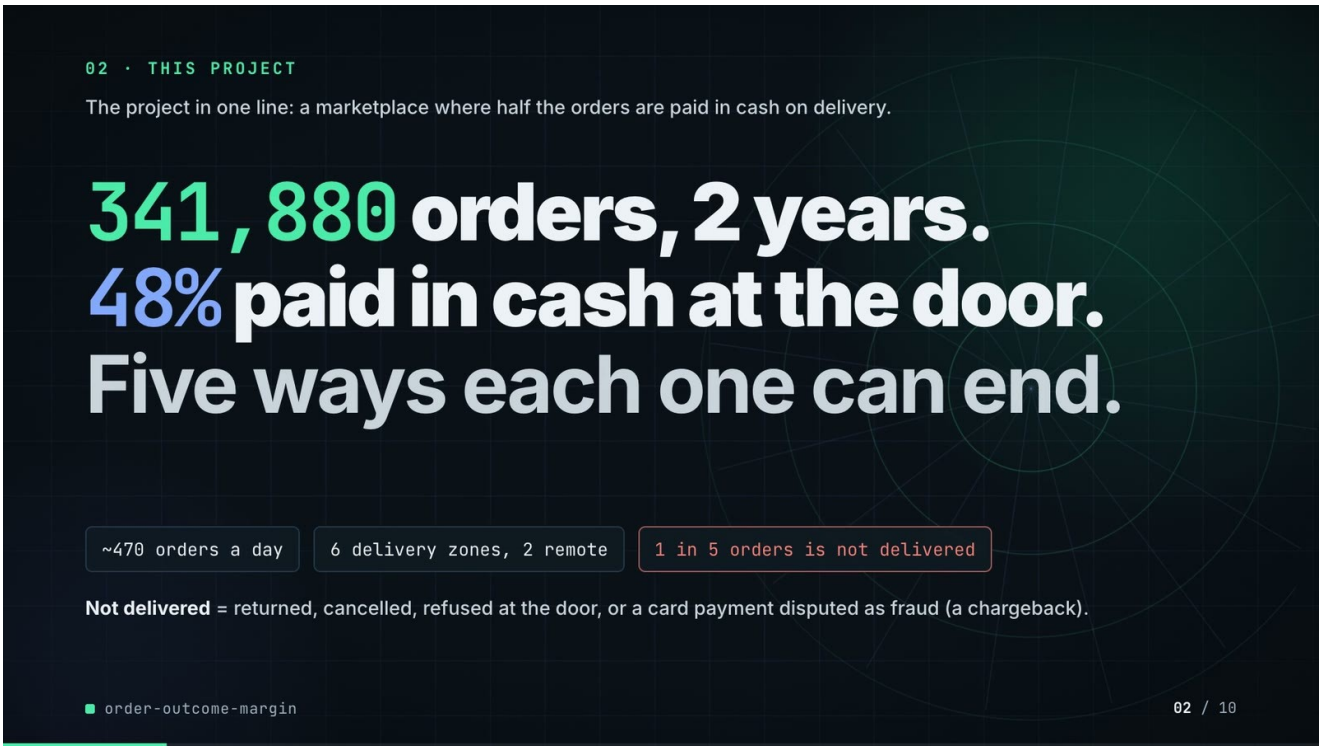
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the subtitle (#ctx-sub), chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-12.0 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

The project in one line: a marketplace where half the orders are paid in cash on delivery.

341,880 orders, 2 years. 48% paid in cash at the door. Five ways each one can end.

~470 orders a day · 6 delivery zones, 2 remote · 1 in 5 orders is not delivered

Not delivered = returned, cancelled, refused at the door, or a card payment disputed as fraud (a chargeback).

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	341,880 orders	`outputs/results.json` · orders_total	-	✓
F-04	2 years / 2024–2025	`src/order_margin/config.py:70-71` · DATA_START, DATA_END	-	✓
F-05	48%	`data/orders.csv.gz` · payment == 'cod'	share of COD orders = 48.3%	✓
F-06	Five ways each one can end / 5 ways it could end	`src/order_margin/config.py:11` · CLASSES	len = 5	✓
F-07	~470 orders a day	`data/orders.csv.gz` · orders / distinct order days	341,880 / 731 = 467.7, rounded to tens (README: 'around 470 orders a day')	✓
F-08	6 delivery zones, 2 remote	`src/order_margin/config.py:17-18` · ZONES, REMOTE_ZONES	-	✓
F-09	1 in 5 orders is not delivered	`data/orders.csv.gz` · outcome != delivered	1 - 0.798 = 20.2% = 1 in 5 (test window: 20.9%)	✓

Assumptions: A-01, A-10

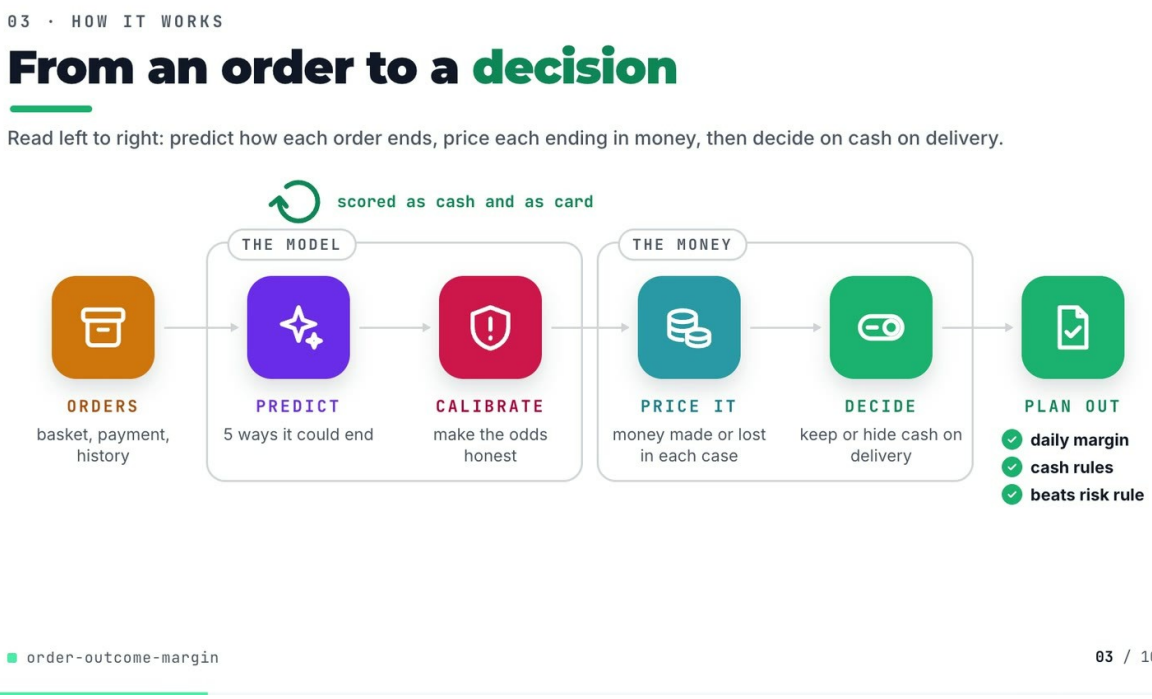
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3, chips #hook-f1..f3 and the definition #hook-def in compositions/s02-hook.html; volumes come from scripts/generate_data.py (make data).

Slide 03: How it works (white chapter, slide A)

11.7-19.7 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From an order to a decision

Read left to right: predict how each order ends, price each ending in money, then decide on cash on delivery.

ORDERS basket, payment, history → [THE MODEL: PREDICT 5 ways it could end → CALIBRATE make the odds honest] → [THE MONEY: PRICE IT money made or lost in each case → DECIDE keep or hide cash on delivery] → PLAN OUT ✓ daily margin ✓ cash rules ✓ beats risk rule

scored as cash and as card (loop over PREDICT)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-06	Five ways each one can end / 5 ways it could end	`src/order_margin/config.py:11` · CLASSES	len = 5	✓
F-10	scored as cash and as card	`src/order_margin/pipeline.py:193-201` · with_payment(Xt_cod, 'card')	-	✓
F-11	daily margin / cash rules / beats risk rule	`outputs/daily_margin.csv`, `outputs/results.json` · 122 daily rows; margin_minus_risk_rule > 0	-	✓
F-12	CALIBRATE · make the odds honest	`src/order_margin/calibration.py:26-39` · IsotonicCalibrator	-	✓

Assumptions: A-02, A-03, A-06

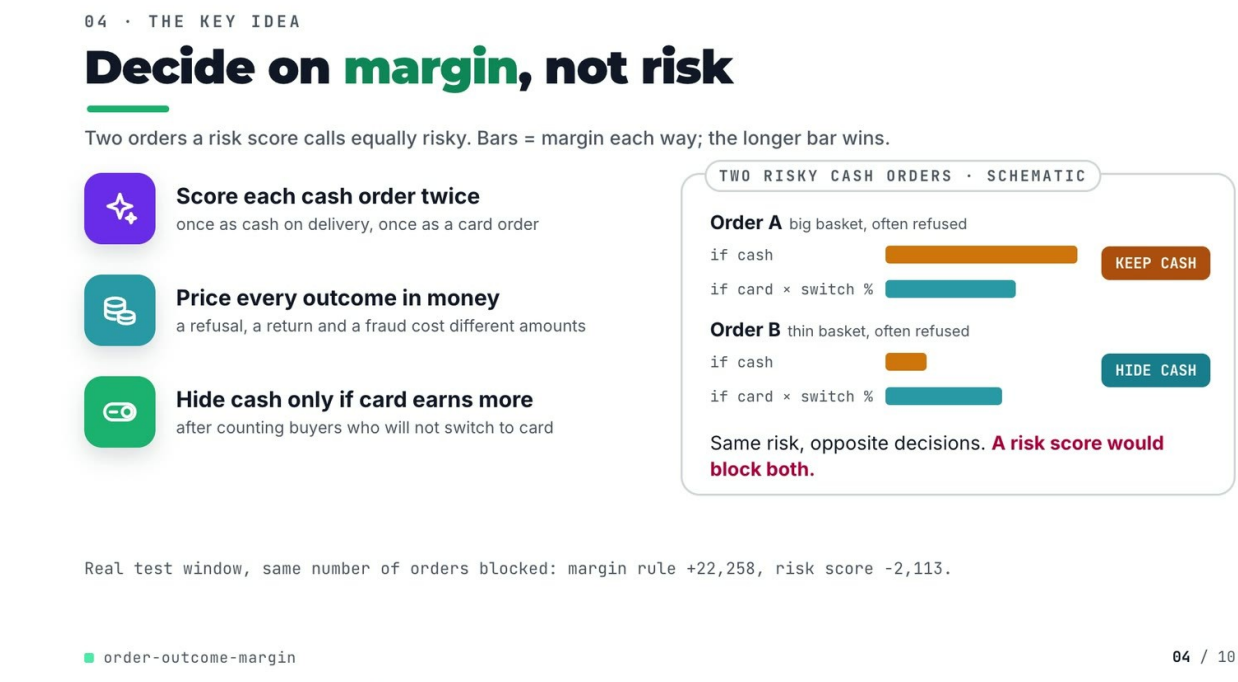
Illustrative, not data:

- Icons and colours are a visual language, not data.

To change this slide: Edit CONFIG and #ex-a-sub in compositions/explainer.html; badge truth comes from F-11.

Slide 04: The key idea (white chapter, slide B)

19.7-26.7 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA

Decide on margin, not risk

Two orders a risk score calls equally risky. Bars = margin each way; the longer bar wins.

Score each cash order twice — once as cash on delivery, once as a card order

Price every outcome in money — a refusal, a return and a fraud cost different amounts

Hide cash only if card earns more — after counting buyers who will not switch to card

TWO RISKY CASH ORDERS · SCHEMATIC: Order A big basket, often refused (if cash > if card × switch %) KEEP CASH; Order B thin basket, often refused (if cash < if card × switch %) HIDE CASH. Same risk, opposite decisions. A risk score would block both.

Real test window, same number of orders blocked: margin rule +22,258, risk score −2,113.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-14	−2,113	`outputs/policy.csv` · policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	✓

Assumptions: A-02, A-03, A-06

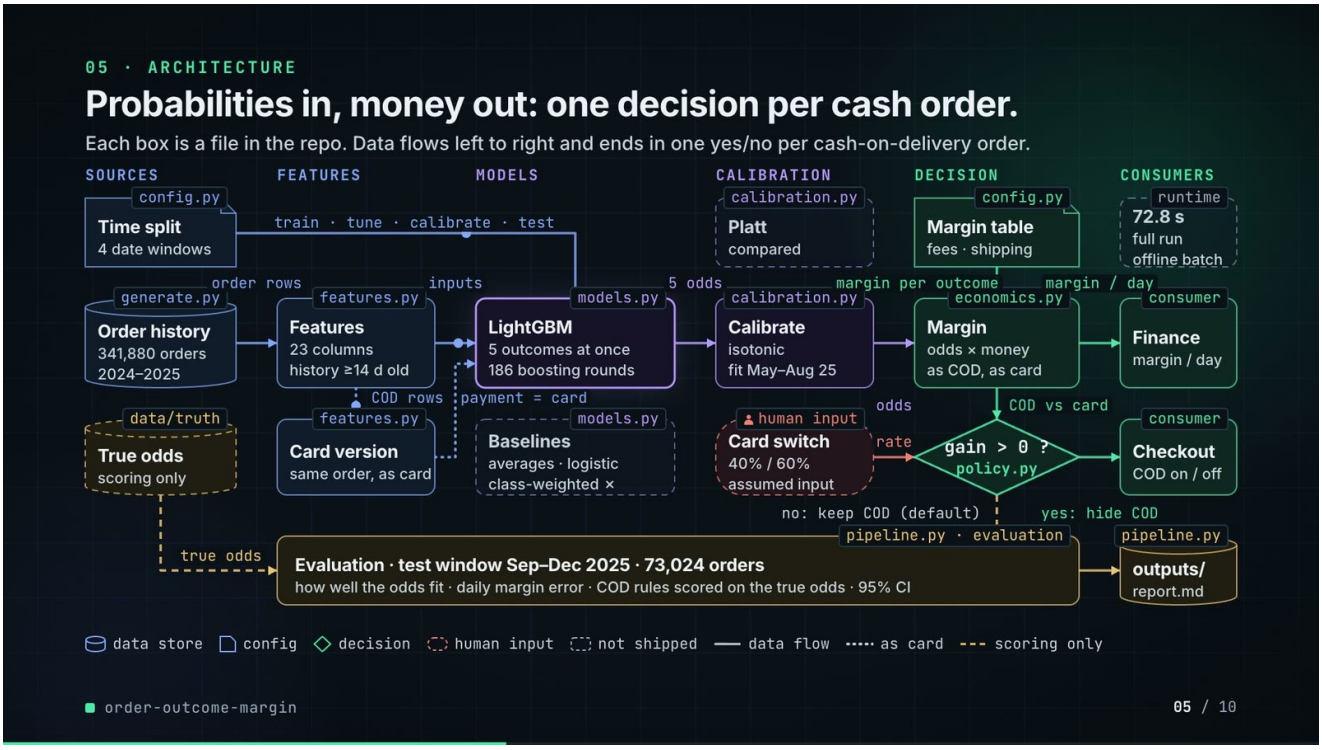
Illustrative, not data:

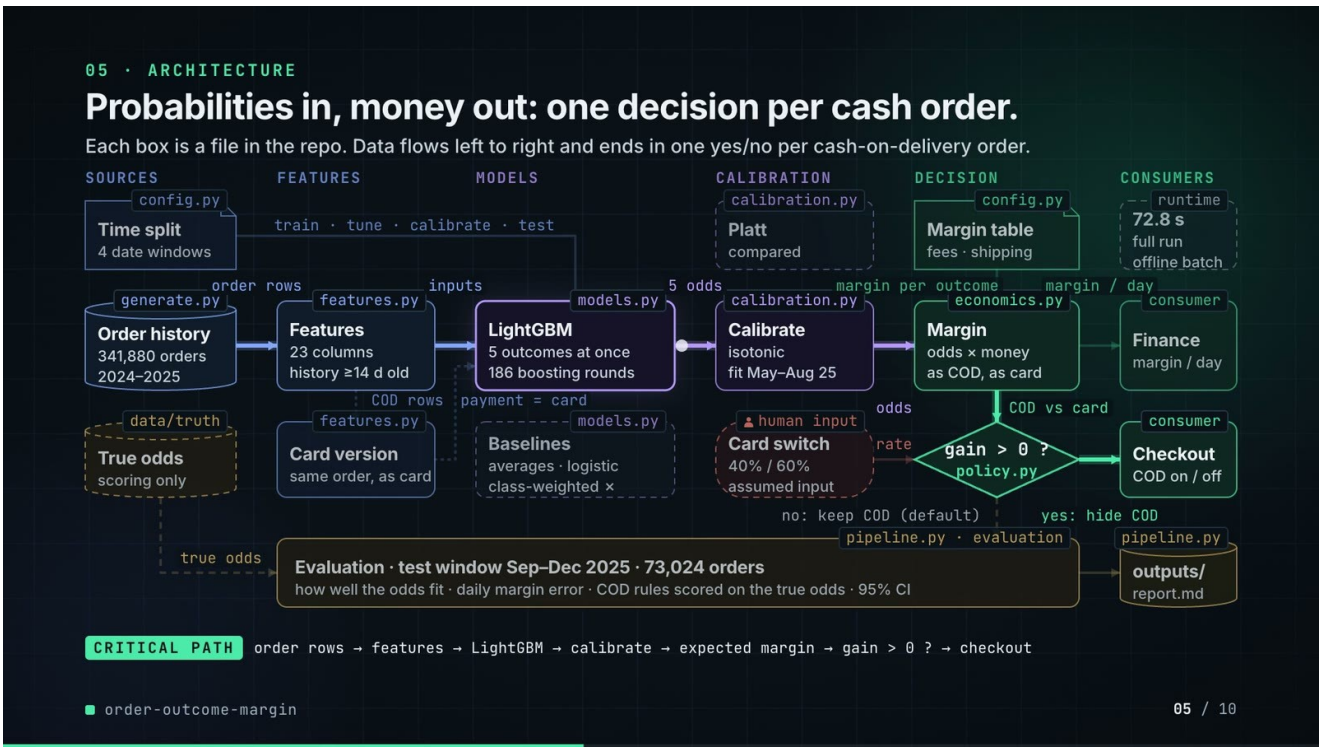
- Orders A and B and their bar lengths are schematic. Real rows with the same pattern are on slide 07 (order 316096: 41% refusal risk, KEEP COD; order 316309: 43%, HIDE COD).

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-13 and F-14.

Slide 05: Architecture

26.4-40.4 s · composition `compositions/s03-architecture.html (N nodes, W wires)`





On-screen text (verbatim)

05 · ARCHITECTURE

Probabilities in, money out: one decision per cash order.

Each box is a file in the repo. Data flows left to right and ends in one yes/no per cash-on-delivery order.

Layers: SOURCES · FEATURES · MODELS · CALIBRATION · DECISION · CONSUMERS

config.py Time split (4 date windows) —train · tune · calibrate · test→ LightGBM

generate.py Order history (341,880 orders, 2024–2025) —order rows→ features.py Features (23 columns, history ≥14 d old) —inputs→ models.py LightGBM (5 outcomes at once, 186 boosting rounds) —5 odds→ calibration.py Calibrate (isotonic, fit May–Aug 25) —odds→ economics.py Margin (odds × money, as COD, as card) —margin / day→ Finance

features.py Card version (same order, as card): COD rows in, payment = card back into LightGBM (dotted, 2nd pass)

config.py Margin table (fees · shipping) —margin per outcome→ Margin; COD vs card → policy.py gain > 0 ? (yes: hide COD → Checkout, COD on / off; no: keep COD (default))

human input: Card switch (40% / 60%, assumed input) —rate→ gain > 0 ?

Ghost boxes (not shipped): Platt (compared), Baselines (averages · logistic, class-weighted ✗); runtime 72.8 s full run, offline batch

data/truth True odds (scoring only) —true odds→ pipeline.py evaluation: Evaluation · test window Sep–Dec 2025 · 73,024 orders · how well the odds fit · daily margin error · COD rules scored on the true odds · 95% CI → outputs/ (report.md)

Legend: data store · config · decision · human input · not shipped · data flow · as card · scoring only

CRITICAL PATH order rows → features → LightGBM → calibrate → expected margin → gain > 0 ? → checkout

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	341,880 orders	`outputs/results.json` · orders_total	-	✓
F-04	2 years / 2024–2025	`src/order_margin/config.py:70-71` · DATA_START, DATA_END	-	✓
F-15	config.py, generate.py, features.py, models.py, calibration.py, economics.py, policy.py, pipeline.py, data/truth	`src/order_margin/` · classes and functions	-	✓
F-16	23 columns	`src/order_margin/features.py:21-45` · NUMERIC (20) + CATEGORICAL (3)	20 + 3 = 23	✓
F-17	history ≥14 d old	`src/order_margin/features.py:15` · RESOLVE_DAYS = 14	-	✓
F-18	186 boosting rounds	`outputs/results.json` · gbm_rounds	-	✓
F-19	5 outcomes at once	`src/order_margin/models.py:74-75` · objective multiclass, num_class = N_CLASSES = 5	-	✓
F-20	fit May–Aug 25	`src/order_margin/config.py:68-69` · EARLY_STOP_END 2025-04-30, CALIB_END 2025-08-31	-	✓
F-21	40% / 60% · assumed input	`src/order_margin/config.py:75` · COD_TO_CARD_CONVERSION	-	✓
F-22	72.8 s	`outputs/results.json` · runtime_seconds	-	✓
F-23	73,024 orders	`outputs/results.json` · orders_test	-	✓
F-24	how well the odds fit · daily margin error · COD rules scored on the true odds · 95% CI	`src/order_margin/pipeline.py:139-160, 274-275; calibration.py:64-83` · -	-	✓
F-25	4 date windows · train · tune · calibrate · test	`src/order_margin/pipeline.py:64-71` · split_masks keys	-	✓

Assumptions: A-01, A-02, A-04, A-05, A-06, A-07

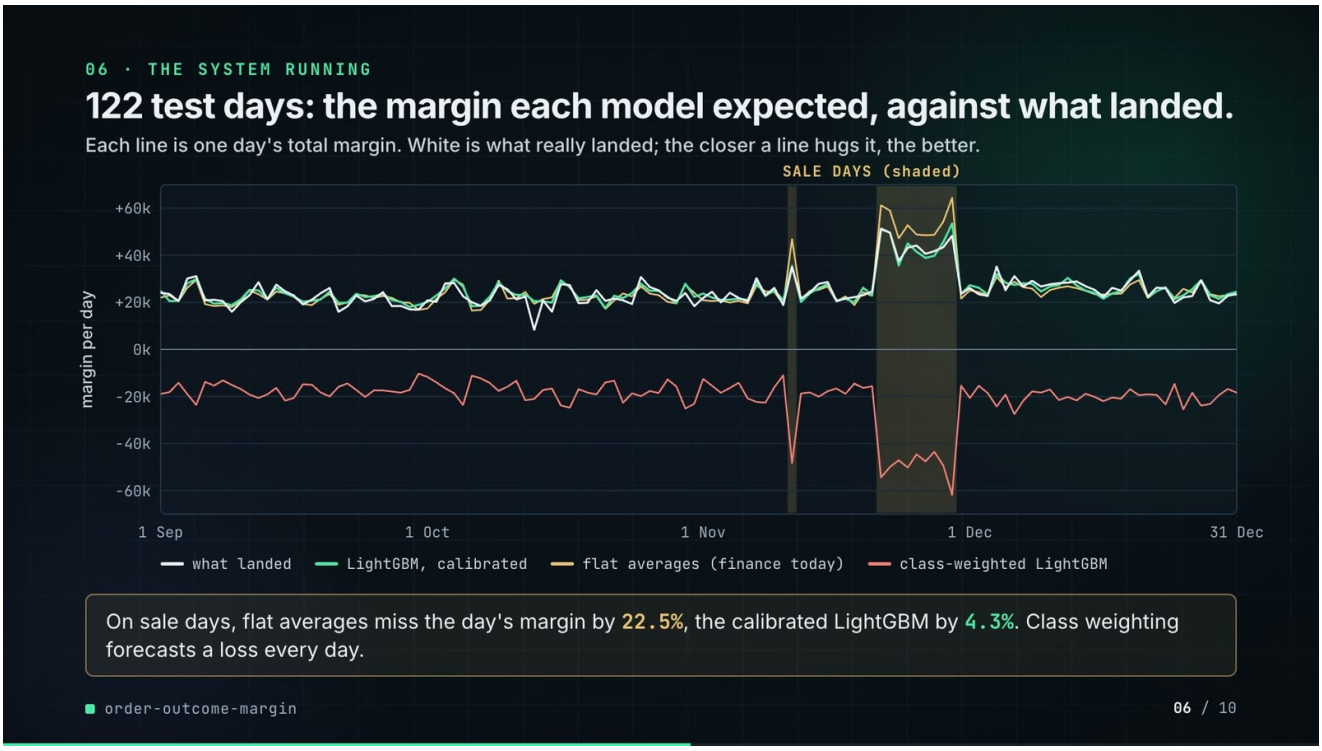
Illustrative, not data:

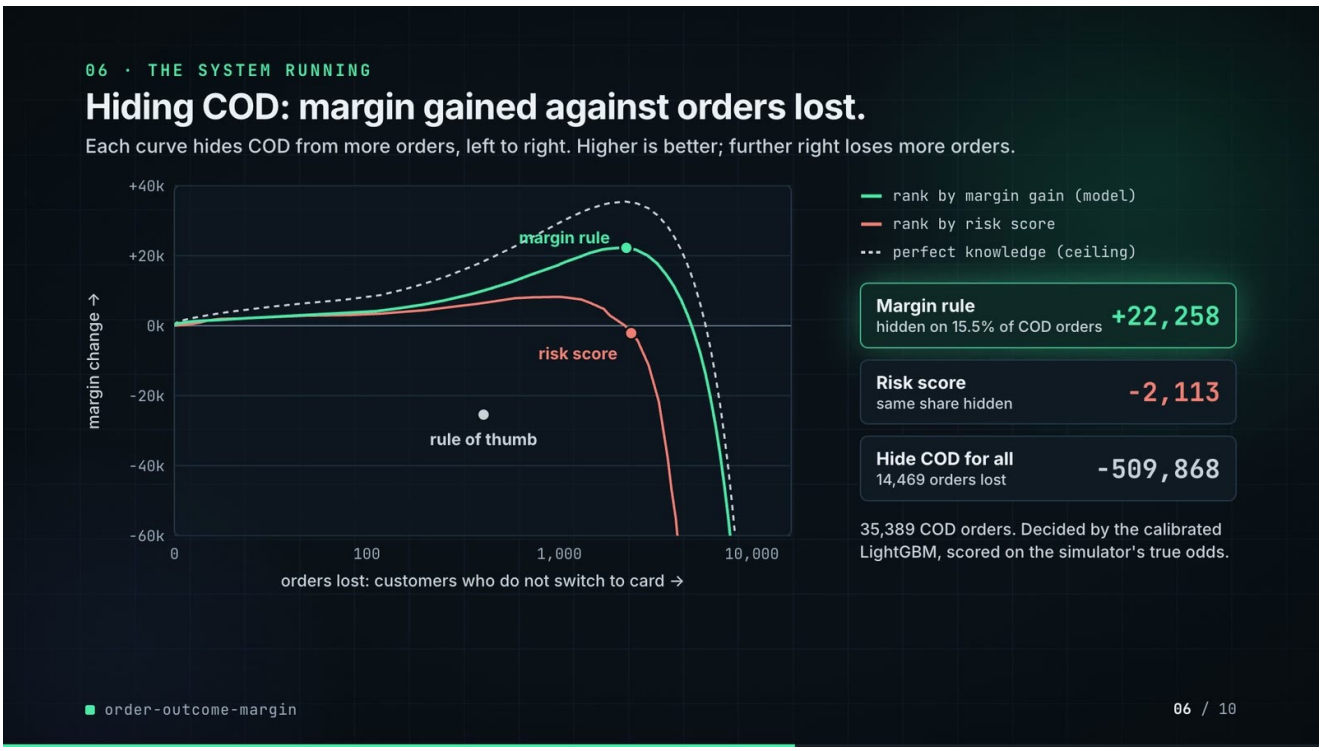
- Packets moving along the wires show execution order, not timing.
- 'Finance' and 'Checkout' are the consumers the README names (finance close, checkout COD option); the repo writes files, it does not call a checkout service.
- 'offline batch' describes how the repo runs (one CLI run over files); no serving latency is measured, so none is shown.
- In the critical-path phase the other boxes dim to 75% opacity (text stays above 4.5:1).

To change this slide: Node array N and wire array W in compositions/s03-architecture.html.

Slide 06: The system running

40.0-54.6 s · composition `compositions/s04-routes.html (RUN block from tools/export_run.py + tools/inject_block.py)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

122 test days: the margin each model expected, against what landed.

Each line is one day's total margin. White is what really landed; the closer a line hugs it, the better.

y: margin per day (−60k … +60k) · x: 1 Sep … 31 Dec · SALE DAYS (shaded)

what landed · LightGBM, calibrated · flat averages (finance today) · class-weighted LightGBM

On sale days, flat averages miss the day's margin by 22.5%, the calibrated LightGBM by 4.3%. Class weighting forecasts a loss every day.

Hiding COD: margin gained against orders lost.

Each curve hides COD from more orders, left to right. Higher is better; further right loses more orders.

x: orders lost: customers who do not switch to card (0, 100, 1,000, 10,000) · y: margin change (−60k … +40k) · points: margin rule, risk score, rule of thumb

rank by margin gain (model) · rank by risk score · perfect knowledge (ceiling)

Margin rule (hidden on 15.5% of COD orders) +22,258 · Risk score (same share hidden) −2,113 · Hide COD for all (14,469 orders lost) −509,868

35,389 COD orders. Decided by the calibrated LightGBM, scored on the simulator's true odds.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-14	−2,113	`outputs/policy.csv` · policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	✓
F-26	SVG line chart	`outputs/daily_margin.csv via tools/export_run.py → assets/run.json → RUN block in s04-routes.html` · realized, class priors, gbm + isotonic, gbm class-weighted, raw, peak	values rounded to whole units; drawn, not retyped	✓
F-27	122 test days	`outputs/results.json` · test_days	-	✓
F-28	22.5%	`outputs/metrics.csv` · model='class priors', daily_mape_peak (0.2254)	x 100	✓
F-29	4.3%	`outputs/metrics.csv` · model='gbm + isotonic', daily_mape_peak (0.0431)	x 100	✓
F-30	Class weighting forecasts a loss every day.	`outputs/daily_margin.csv` · column 'gbm class-weighted, raw' max = -10,259	max < 0	✓
F-31	SVG line chart, y clipped to -60k..+40k	`outputs/policy_frontier.csv`, `outputs/policy.csv via tools/export_run.py` · orders_lost, margin_change per ranking; policy rows	same symlog x-axis as plots.py (linear below 100)	✓
F-32	−509,868	`outputs/policy.csv` · policy='block every COD order', margin_change	-	✓
F-33	14,469 orders lost	`outputs/policy.csv` · orders_lost = 14,468.6	rounded	✓
F-34	hidden on 15.5% of COD orders	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', blocked_share_of_cod (0.1554)	x 100	✓
F-35	35,389 COD orders	`outputs/results.json` · cod_orders_test	-	✓
F-36	Risk score · same block rate	`src/order_margin/pipeline.py:206-218` · threshold_for_share(risk_score(pc_c), share)	-	✓

F-31 note: Curves leave the bottom of the chart; 'hide COD for all' (-509,868) is off the chart and shown in the panel.

Assumptions: A-02, A-03, A-04, A-06, A-09

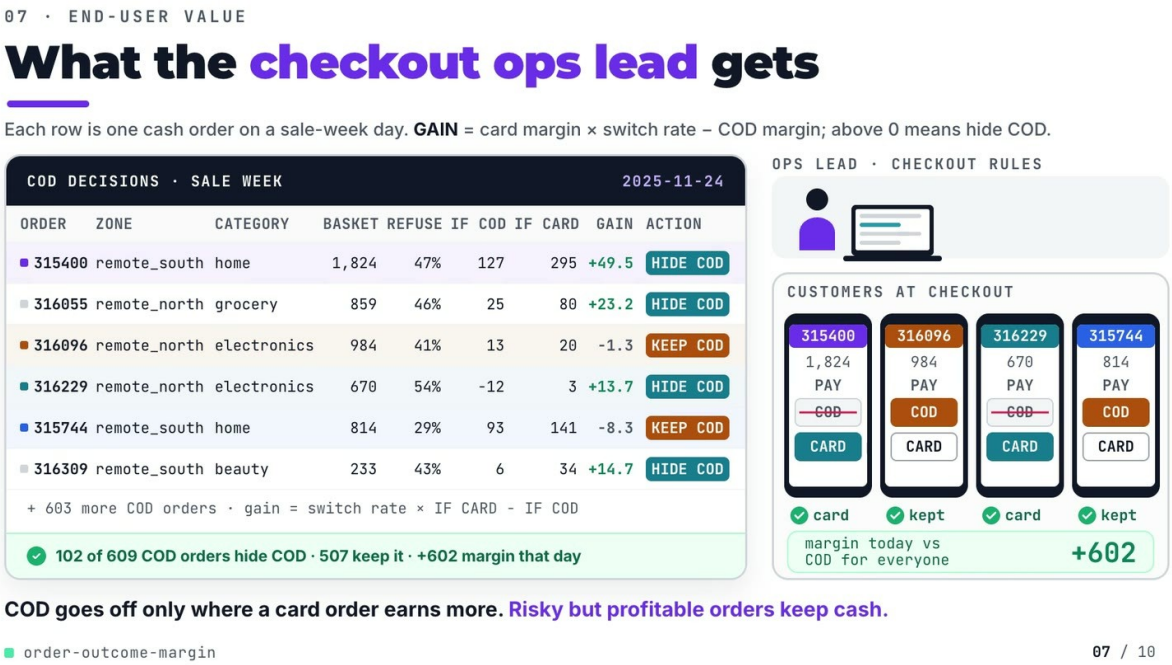
Illustrative, not data:

- Both charts are drawn as SVG from the repo's CSVs (F-26, F-31); no value is typed by hand.
- The frontier's y axis is clipped to −60k…+40k so the useful top band is readable; curves leave the bottom, and 'hide COD for all' is reported in the panel instead.

To change this slide: Re-run tools/export_run.py and tools/inject_block.py after make run; captions and panel text in compositions/s04-routes.html.

Slide 07: End-user value (white)

54.3-67.3 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the checkout ops lead gets

Each row is one cash order on a sale-week day. GAIN = card margin × switch rate − COD margin; above 0 means hide COD.

COD DECISIONS · SALE WEEK · 2025-11-24 — columns ORDER, ZONE, CATEGORY, BASKET, REFUSE, IF COD, IF CARD, GAIN, ACTION

315400 remote_south home 1,824 47% 127 295 +49.5 HIDE COD

316055 remote_north grocery 859 46% 25 80 +23.2 HIDE COD

316096 remote_north electronics 984 41% 13 20 −1.3 KEEP COD

316229 remote_north electronics 670 54% −12 3 +13.7 HIDE COD

315744 remote_south home 814 29% 93 141 −8.3 KEEP COD

316309 remote_south beauty 233 43% 6 34 +14.7 HIDE COD

+ 603 more COD orders · gain = switch rate × IF CARD − IF COD · ✓ 102 of 609 COD orders hide COD · 507 keep it · +602 margin that day

OPS LEAD · CHECKOUT RULES · CUSTOMERS AT CHECKOUT (4 phones: order id, basket, PAY COD / CARD) · card / kept · margin today vs COD for everyone +602

COD goes off only where a card order earns more. Risky but profitable orders keep cash.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-37	SALE WEEK · 2025-11-24	`demo assets/cod_sheet.json; src/order_margin/generate.py:30` · day, is_peak	-	✓
F-38	609 COD orders (6 rows + 603 more)	`demo assets/cod_sheet.json; data/orders.csv.gz` · cod_orders; COD orders placed on 2025-11-24	-	✓
F-39	102 of 609 COD orders hide COD · 507 keep it	`demo assets/cod_sheet.json` · blocked, kept	102 + 507 = 609	✓
F-40	+602 margin that day	`demo assets/cod_sheet.json` · day_true_gain	sum over hidden orders of conv x E_true[card] - E_true[cod]	✓
F-41	315400, 316055, 316096, 316229, 315744, 316309	`demo assets/cod_sheet.json, injected into compositions/value.html` · rows	gain = conv x IF CARD - IF COD; action = HIDE COD if gain > 0	✓
F-42	(consistency check, not on screen)	`demo assets/cod_sheet.json; outputs/policy.csv` · test_window_blocked vs policy 'margin rule, gbm + isotonic' cod_orders_blocked	5,498 = 5,498; 186 rounds = 186	✓
F-43	+ 603 more COD orders	`F-38` · 609 - 6 rows	609 - 6 = 603	✓

Assumptions: A-02, A-03, A-06, A-07

Illustrative, not data:

- Ops-lead figure, laptop, phones and tokens are illustrative; no real people.
- The four phones show four of the six rows (two hidden, two kept), chosen to show both decisions. Order ids are the generator's synthetic ids.
- 'card' under a phone means the order is offered card only; whether that customer converts is the 40/60% assumption (A-02).
- IF COD / IF CARD are the model's expected margins; the day total (+602) is scored on true probabilities, like the repo's policy table.

To change this slide: Re-run tools/export_cod_sheet.py, then tools/inject_sheet.py (--rows / --animate pick the orders); layout in compositions/value.html.

Slide 08: Results

67.0-75.8 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · TEST WINDOW SEP–DEC 2025

Margin earned on COD, and a forecast that holds in sale week.

Margin = profit after shipping, fees and losses, so higher is better. Forecast error: lower is better.

DECISION +22,258 extra margin from the COD rule · risk score: −2,113

DECISION 63% of the gain a perfect model gets · perfect model: +35,479

FORECAST 4.3% forecast error per sale day · flat averages: 22.5%

EVALUATION +24,371 better than ranking COD orders on risk · 95% CI: 21,580–27,556

73,024 test orders, 35,389 paid by COD. Every rule is scored on the simulator's true odds. CI = 95% confidence range.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-14	−2,113	`outputs/policy.csv` · policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	✓
F-23	73,024 orders	`outputs/results.json` · orders_test	-	✓
F-28	22.5%	`outputs/metrics.csv` · model='class priors', daily_mape_peak (0.2254)	x 100	✓
F-29	4.3%	`outputs/metrics.csv` · model='gbm + isotonic', daily_mape_peak (0.0431)	x 100	✓
F-35	35,389 COD orders	`outputs/results.json` · cod_orders_test	-	✓
F-44	63%	`outputs/policy.csv` · margin rule 22,258 / true-probability ceiling 35,479	22,258 / 35,479 = 62.7%	✓
F-45	perfect model: +35,479	`outputs/policy.csv` · policy='margin rule, true probabilities (ceiling)', margin_change	-	✓
F-46	+24,371	`outputs/results.json` · margin_minus_risk_rule	-	✓
F-47	95% CI: 21,580–27,556	`outputs/results.json` · margin_minus_risk_rule_ci95	-	✓

Assumptions: A-02, A-03, A-06, A-08, A-09, A-10

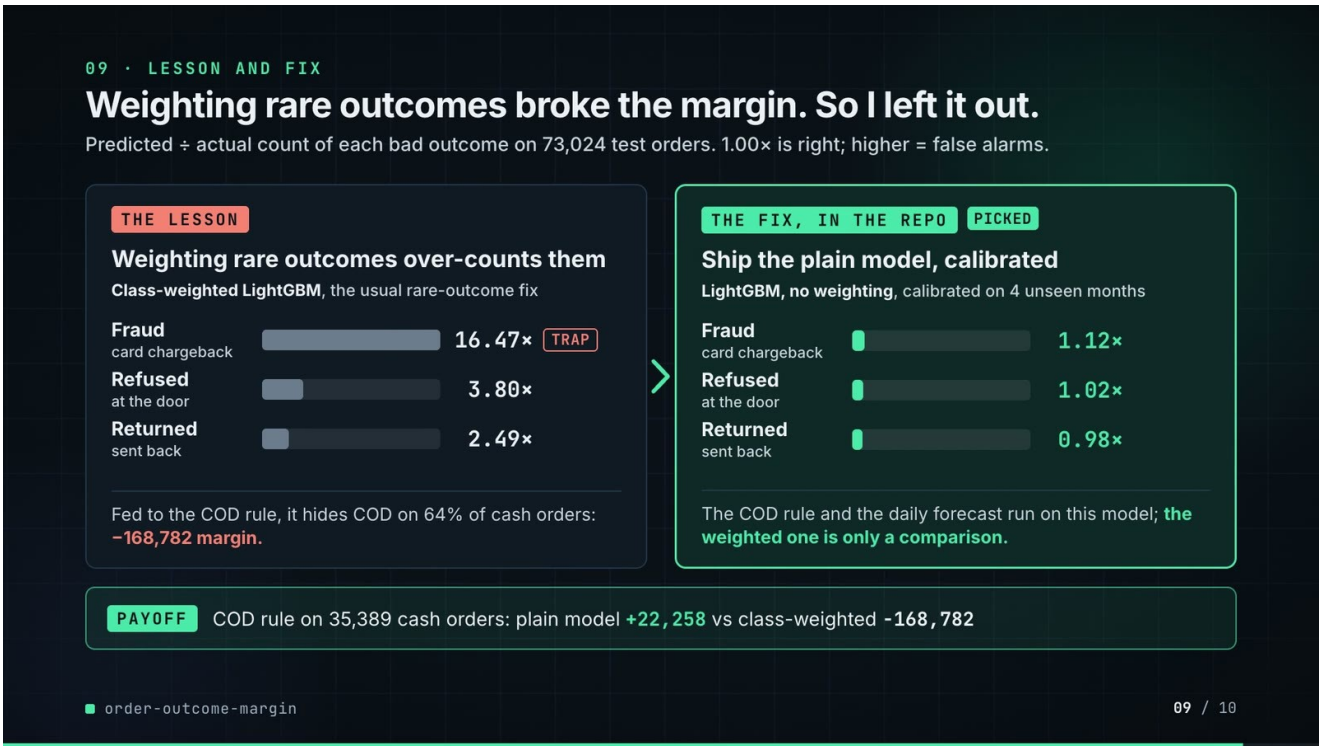
Illustrative, not data:

- Mini bars: cards 1-2 = 22,258 / 35,479; card 3 = 4.3 / 22.5; card 4 is a full bar (no ratio).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: Lesson and fix

75.4-84.4 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · LESSON AND FIX

Weighting rare outcomes broke the margin. So I left it out.

Predicted ÷ actual count of each bad outcome on 73,024 test orders. 1.00× is right; higher = false alarms.

THE LESSON · Weighting rare outcomes over-counts them · Class-weighted LightGBM, the usual rare-outcome fix · Fraud (card chargeback) 16.47× TRAP · Refused (at the door) 3.80× · Returned (sent back) 2.49×

Fed to the COD rule, it hides COD on 64% of cash orders: −168,782 margin.

THE FIX, IN THE REPO · PICKED · Ship the plain model, calibrated · LightGBM, no weighting, calibrated on 4 unseen months · Fraud 1.12× · Refused 1.02× · Returned 0.98×

The COD rule and the daily forecast run on this model; the weighted one is only a comparison.

PAYOFF · COD rule on 35,389 cash orders: plain model +22,258 vs class-weighted −168,782

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-23	73,024 orders	`outputs/results.json` · orders_test	-	✓
F-35	35,389 COD orders	`outputs/results.json` · cod_orders_test	-	✓
F-48	16.47×	`outputs/metrics.csv` · model='gbm class-weighted, raw', count_ratio_fraud	-	✓
F-49	3.80×	`outputs/metrics.csv` · model='gbm class-weighted, raw', count_ratio_failed	-	✓
F-50	2.49×	`outputs/metrics.csv` · model='gbm class-weighted, raw', count_ratio_returned	-	✓
F-51	1.12×	`outputs/metrics.csv` · model='gbm + isotonic', count_ratio_fraud	-	✓
F-52	1.02×	`outputs/metrics.csv` · model='gbm + isotonic', count_ratio_failed	-	✓
F-53	0.98×	`outputs/metrics.csv` · model='gbm + isotonic', count_ratio_returned	-	✓
F-54	64% of cash orders	`outputs/policy.csv` · policy='margin rule, class-weighted gbm', blocked_share_of_cod (0.6393)	x 100	✓
F-55	−168,782	`outputs/policy.csv` · policy='margin rule, class-weighted gbm', margin_change	-	✓
F-56	bars 260 / 60 / 39 px (lesson) and 18 / 16 / 15 px (fix)	`outputs/metrics.csv` · count_ratio_* / max x 260	round(ratio / 16.4749 x 260)	✓
F-61	Ship the plain model, calibrated · PICKED · the	`src/order_margin/models.py:71; pipeline.py:122, 180, 203, 229, 252; tests/test_pipeline.py:24` · -	-	✓

ID	Shown	Source · key	Calculation	Check
	weighted one is only a comparison			

Assumptions: A-02, A-03, A-06, A-07, A-10, A-12

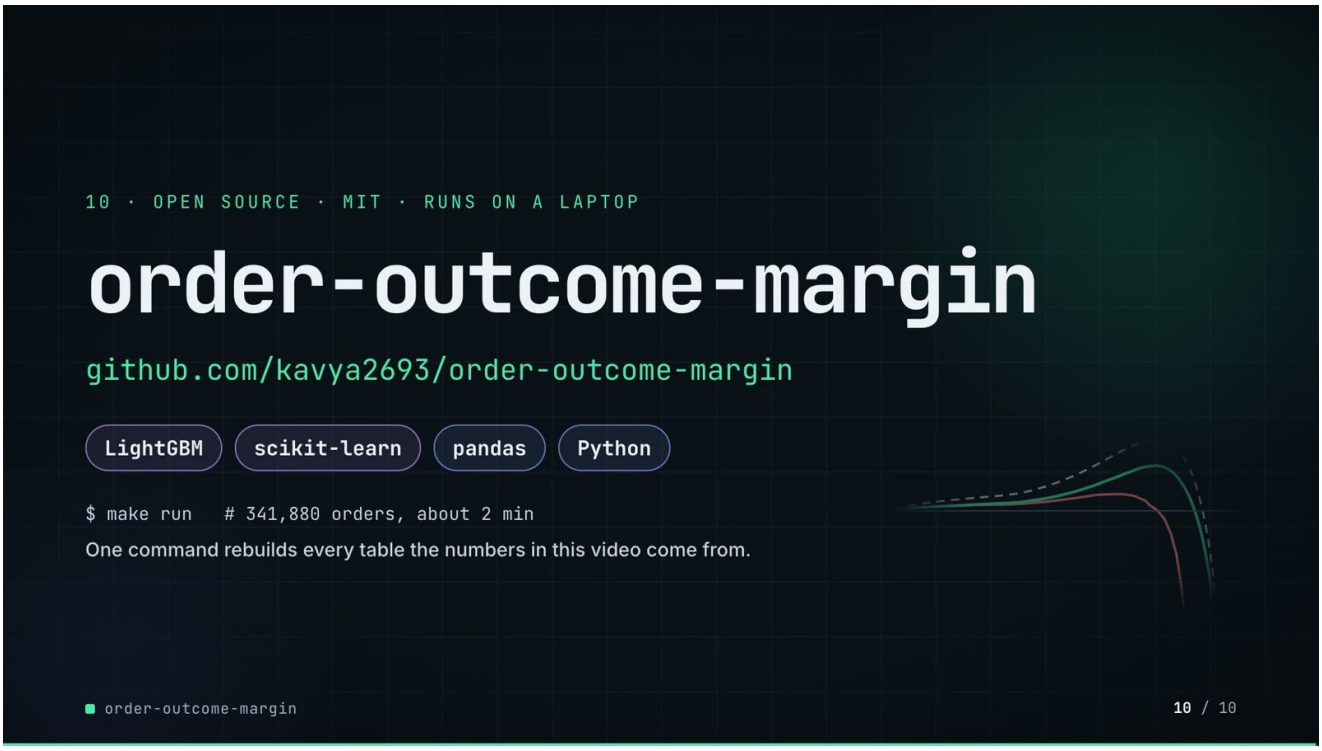
Illustrative, not data:

- Bars use one shared scale, 16.47× = 260 px (F-56); bars are neutral grey on the lesson side and mint for the shipped model.

To change this slide: Rows #twist-l1..l3 / #twist-r1..r3 (bar widths are inline styles) and the payoff strip in compositions/s06-twist.html.

Slide 10: End card

84.0-88.4 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP
order-outcome-margin
github.com/kavya2693/order-outcome-margin
LightGBM · scikit-learn · pandas · Python
$ make run # 341,880 orders, about 2 min
One command rebuilds every table the numbers in this video come from.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-57	MIT	`LICENSE line 1` · -	-	✓
F-58	about 2 min	`README.md line 137` · 'about 2 minutes on a laptop'	-	✓
F-59	LightGBM · scikit-learn · pandas · Python	`pyproject.toml dependencies` · lightgbm, scikit-learn, pandas	-	✓
F-60	$ make run	`Makefile` · run target	-	✓

Assumptions: A-10

Illustrative, not data:

- Faded curves in the corner are the COD frontier from outputs/policy_frontier.csv, drawn without text by tools/make_end_plot.py.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (341,880 orders over 2024-2025, six zones, five categories, COD about 48%)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/order_margin/generate.py`; `src/order_margin/world.py`	make data; make run
A-02: Card conversion: 40% of blocked customers younger than 30 days and 60% of the rest switch to a card	F-13, F-14, F-21, F-32, F-33, F-34, F-39, F-40, F-41, F-44, F-45, F-46, F-47, F-54, F-55	04, 05, 06, 07, 08, 09	`src/order_margin/config.py:75`	make run; export_cod_sheet.py + inject_sheet.py
A-03: Unit economics are one table per category (gross margin, resale loss, shipping legs, COD/card/wallet fees, a 45 chargeback fee)	F-13, F-14, F-28, F-29, F-30, F-32, F-40, F-41, F-44, F-45, F-46, F-47, F-55	04, 06, 07, 08, 09	`src/order_margin/config.py:22-60`	make run; export_cod_sheet.py + inject_sheet.py
A-04: Time split: train 2024-01-01	F-20, F-23, F-25, F-27, F-28, F-29, F-35, F-48, F-49, F-50, F-51, F-52, F-53	05, 06, 08, 09	`src/order_margin/config.py:67-71`	make run
A-05: Outcomes are known 14 days after the order; customer history features only use orders placed at least 14 days earlier, shrunk towards fixed long-run rates	F-17	05	`src/order_margin/features.py:15`	make run
A-06: COD policies are scored on the generator's true outcome probabilities under COD and under card (data/truth/order_truth	F-13, F-14, F-32, F-40, F-44, F-45, F-46, F-47, F-55	04, 06, 07, 08, 09	`src/order_margin/pipeline.py`	make run
A-07: LightGBM settings: multiclass, learning rate 0	F-18, F-19, F-48, F-49, F-50, F-51, F-52, F-53	05, 09	`src/order_margin/models.py:71-89`	make run
A-08: One seed and one test window	F-46, F-47	08	`src/order_margin/config.py:78`	make data; make run
A-09: 'Sale days' = days flagged is_peak: the November sale week (2025-11-21	F-28, F-29, F-37, F-26	06, 07, 08	`src/order_margin/generate.py:30-31`	make data; make run
A-10: Video rounding: margins to whole units with thousands separators, percentages to 1 decimal (63%, 48% and 64% to whole numbers), predicted/actual count ratios to 2 decimals; chart axes in thousands ('+20k'); '~470 a day' and 'about 2 min' are rounded labels	F-05, F-07, F-09, F-22, F-28, F-29, F-34, F-44, F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-58	02, 05, 06, 08, 09, 10	the composition file named on each slide	hyperframes render
A-11: Slide 01 (industry, everyday problems) is general e-commerce knowledge used as context, matched to the README problem statement	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render
A-12: The shipped model is the plain (unweighted) LightGBM with an isotonic calibrator fitted on May-Aug 2025, a window the booster never trained or early-stopped on	F-48, F-49, F-50, F-51, F-52, F-53, F-54, F-55, F-61	09	`src/order_margin/pipeline.py:203`; `src/order_margin/models.py:71`	make run

6. Known differences and caveats

• The COD gain rests on an assumed card conversion (40% / 60%). With decisions fixed, 20 points lower conversion breaks even and 30 points lower loses 9,714 (outputs/conversion_sensitivity.csv, A-02). The README names this the biggest gap.

• Policies are scored on the generator's true probabilities, which only exist because the data is synthetic (A-06). On real data the same comparison needs a randomised checkout holdout.

• Orders lost are counted, not priced: a customer pushed away may not come back (README 'Known limitations').

• 'Sale days' in the 22.5% / 4.3% figures include 11.11 as well as the Nov 21-29 sale week; slide 06 shades all ten days (A-09).

• Slide 09's payoff compares the class-weighted model without calibration (the repo's 'margin rule, class-weighted gbm') against the shipped LightGBM + isotonic. The plain booster without calibration scores +22,405 (outputs/policy.csv 'margin rule, uncalibrated gbm'), so the gap comes from the weighting, not from calibration.

• Slide 09 no longer shows two narrower results the README still reports under 'What didn't win' and 'The result': LightGBM barely beats logistic regression (log-loss 0.6469 vs 0.6475), and isotonic calibration did nothing for the plain booster (ECE 0.0027 -> 0.0035). Isotonic did repair the class-weighted model (ECE 0.206 -> 0.009) but its margin stayed 4.2% off.

• Slide 07 rows come from a re-run of the repo's own model for 2025-11-24 (tools/export_cod_sheet.py); the re-run reproduces the repo's 186 rounds and its 5,498 hidden COD orders over the test window exactly (F-42).

• The architecture's 'Finance' and 'Checkout' are the README's consumers; the repo itself writes files (outputs/) and runs as one offline batch. No serving latency exists, so none is shown.

• Slide 01 statements are general industry knowledge (A-11).

Appendix: environment

Item	Value
Commit	`e78d53a391889af780be9f07f440398e04edc426`
Python	3.11.7
lightgbm	4.7.0
sklearn	1.9.1
pandas	3.0.6
numpy	2.4.6
Profile / seeds	data_seed=20260913, lightgbm_seed=7, bootstrap_seed=0
Run time	results.json runtime_seconds = 72.8 s for the pipeline (make run also generates the data first; README says about 2 minutes on a laptop)